

INTERFERENCE

26.1. PRINCIPLE OF SUPERPOSITION

This principle states that the resultant displacement of a particle of the medium acted upon by two or more waves simultaneously is the algebraic sum of the displacements of the same particle due to individual waves, in the absence of the others.

Suppose due to a single wave train the displacement of the particle at a certain point at any instant is y_1 in a given direction and that due to another wave train, the displacement is y_2 in the absence of the first. According to the principle of superposition, the instantaneous resultant displacement of the particle due to two waves acting together is expressed by

$$R_1 = y_1 + y_2$$

If the two displacements are in opposite directions, the instantaneous resultant displacement due to two waves acting together is expressed by

$$R_2 = y_1 - y_2$$

26.2. INTERFERENCE OF LIGHT

When two light waves superimpose, then the resultant amplitude (or intensity*) in the region of superposition is different than the amplitude (or intensity) of individual waves. This modification in the distribution of intensity in the region of superposition is called interference. When the resultant amplitude is the sum of the amplitudes due to two waves, the interference is known as constructive interference and when the resultant amplitude is equal to the difference of two amplitudes, the interference is known as destructive interference.

26.3. YOUNG'S EXPERIMENT

In 1801, Thomas Young first demonstrated experimentally the phenomenon of interference of light. The apparatus is shown in Fig. 26.1. He al-

lowed the sunlight to pass through a pin hole S and then at some distances through two sufficiently close pin holes S_1 and S_2 in an opaque screen. The interference pattern was observed on a screen XY . He observed few coloured bright and dark bands on the screen. Now a days, to increase the brightness, pin holes S_1 and S_2 are replaced by narrow slits, moreover, to increase the number of fringes, sunlight is replaced by monochromatic light. The interference pattern consists of equally spaced bright and dark bands.

Explanation on wave theory. When sunlight passes through pin hole S , spherical waves are

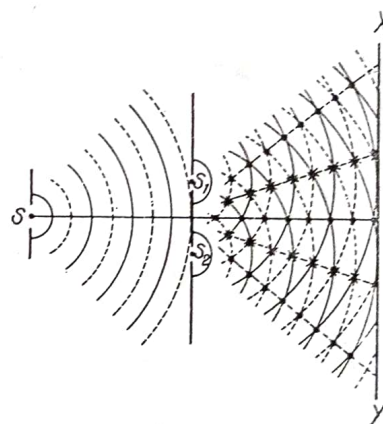


Fig. 26.1

spread out. We know by Huygens' principle that each point on the wavefront is a centre of secondary wavelets. Hence spherical waves also spread out from pin holes S_1 and S_2 . The radii of these wavefronts increase as they move away from S_1 and S_2 and hence they superimpose on each other.

In Fig. 26.1, continuous circular arcs represent the wave crests while dotted circular arcs represent the wave troughs in each wave. At points where a

* Intensity is directly proportional to the square of the amplitude.

crest due to one wave falls on the crest due to another wave (or trough on a trough), the resultant amplitude is the sum of the amplitudes due to each wave separately. As the intensity is proportional to the square of the amplitude and hence the resultant intensity at these points is increased. This is known as constructive interference. At points, where a crest due to one wave falls on the trough of the other wave, the resultant amplitude is equal to the difference of the amplitudes due to separate waves and hence the resultant intensity is minimum. This is known as destructive interference. Thus on the screen a number of alternate bright and dark regions of equal width, called interference fringes, are observed.

26.4. ANALYTICAL TREATMENT OF INTERFERENCE

Consider a monochromatic source of light S emitting waves and let S_1 and S_2 be two narrow pin holes (Fig. 26.2) equidistant from the source S . The

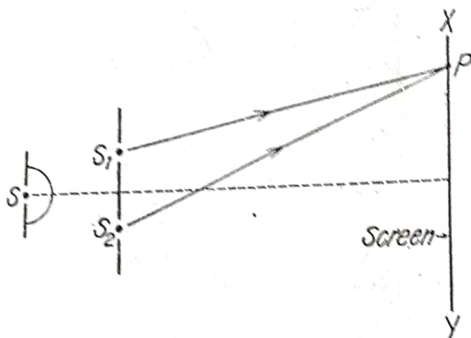


Fig. 26.2

waves arriving at S_1 and S_2 from S will be in phase at all times. We shall investigate the resultant intensity of light at a point P of the screen XY placed parallel to S_1 and S_2 as a result of interference. Let a_1 and a_2 be the amplitude of the waves from S_1 and S_2 and δ be the phase difference between the two waves reaching at point P . If y_1 and y_2 be the displacements of the two waves, then

$$y_1 = a_1 \sin \omega t \quad \dots(1)$$

$$\text{and } y_2 = a_2 (\sin \omega t + \delta) \quad \dots(2)$$

Now according to the principle of superposition, when a particle is disturbed simultaneously by two or more waves, the resultant displacement is equal to the sum of the displacements due to each wave separately. Hence the resultant displacement

$$y = y_1 + y_2 = a_1 \sin \omega t + a_2 \sin (\omega t + \delta)$$

$$\begin{aligned} &= a_1 \sin \omega t + a_2 \sin \omega t \cos \delta + a_2 \cos \omega t \sin \delta \\ &= \sin \omega t (a_1 + a_2 \cos \delta) + \cos \omega t (a_2 \sin \delta) \end{aligned} \quad \dots(3)$$

$$\text{Let } a_1 + a_2 \cos \delta = R \cos \theta \quad \dots(4)$$

$$\text{and } a_2 \sin \delta = R \sin \theta \quad \dots(5)$$

where R and θ are new constants.

Substituting Eqs. (4) and (5) in Eq. (3), we get

$$y = \sin \omega t \cdot R \cos \theta + \cos \omega t \cdot R \sin \theta$$

$$\text{or } y = R \sin (\omega t + \theta) \quad \dots(6)$$

Hence the resultant vibration at P is simple harmonic vibration of amplitude R . The resultant amplitude can be obtained by squaring equations (4) and (5) and adding. Thus we have

$$R^2 \cos^2 \theta + R^2 \sin^2 \theta = (a_1 + a_2 \cos \delta)^2 + (a_2 \sin \delta)^2$$

$$\begin{aligned} \text{or } R^2 &= a_1^2 + a_2^2 \cos^2 \delta + 2a_1 a_2 \cos \delta + a_2^2 \sin^2 \delta \\ &= a_1^2 + a_2^2 + 2a_1 a_2 \cos \delta \end{aligned} \quad \dots(7)$$

The resultant intensity at P is given by the square of the amplitude R . Thus

$$I = R^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos \delta \quad \dots(8)$$

The phase difference δ between the two waves arriving at P is given by

$$\delta = \frac{2\pi}{\lambda} (\text{Path difference})$$

$$= \frac{2\pi}{\lambda} (S_2P - S_1P) \quad \dots(9)$$

Now we consider the following two cases:

Case (1) : Condition for maximum intensity :

Intensity I is maximum, when $\cos \delta = +1$, that is Phase difference

$$\delta = 2n\pi, n = 0, 1, 2, 3, \dots \text{etc.}$$

$$\text{or } \delta = 0, 2\pi, 4\pi, \dots \quad \dots(10)$$

$$\text{or Path difference } (S_2P - S_1P) = n\lambda \quad \dots(11)$$

$$\text{Now } I_{\max} = a_1^2 + a_2^2 + 2a_1 a_2 = (a_1 + a_2)^2 \quad \dots(12)$$

This shows that the resultant intensity is greater than the sum of two separate intensities ($a_1^2 + a_2^2$).

Case (2) : Condition for minimum intensity :

The intensity I is minimum when $\cos \delta = -1$, that is

$$\text{Phase difference } \delta = (2n + 1)\pi, n = 0, 1, 2, 3, \dots \text{etc.}$$

$$\delta = \pi, 3\pi, 5\pi, \dots \text{etc} \quad \dots(13)$$

or Path difference = $(S_2P - S_1P) = (2n + 1) \lambda / 2$... (14)

Now $I_{\min} = a_1^2 + a_2^2 - 2a_1a_2 = (a_1 - a_2)^2$... (15)

Thus the resultant intensity is less than the sum of the two separate intensities $(a_1^2 + a_2^2)$.

Special case : When $a_1 = a_2$, then

$$I_{\max} = a^2 + a^2 + 2a^2 = 4a^2 \quad \dots (16)$$

(where $a_1 = a_2 = a$)

and $I_{\min} = a^2 + a^2 - 2a^2 = 0$... (17)

Energy distribution. From equation (16) it is clear that the intensity at bright points is $4a^2$ while from equation (17) the intensity at dark points is zero. Figure 26.3 shows the variation of intensity as

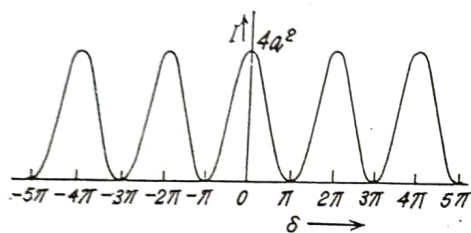


Fig. 26.3

a function δ . The formation of interference fringes is in accordance with the law of conservation of energy. There is no destruction of energy as a result of interference. The energy is simply redistributed. The intensity at maxima increases at the expense of the intensity of minima.

26.5. THEORY OF INTERFERENCE FRINGES

Consider a narrow monochromatic source S and two pin holes S_1 and S_2 equidistant from S . The pin holes provide the two point sources vibrating in the same phase. The two sources are separated at a distance $2d$. Let a screen XY [Fig. 26.4] be placed at a distance D parallel to S_1S_2 . The point O on the screen is equidistance from S_1 and S_2 and hence the path difference between the two waves reaching at O from S_1 and S_2 is zero. Therefore at this

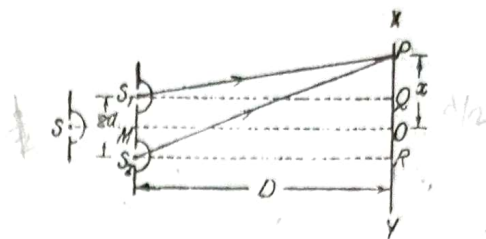


Fig. 26.4

point intensity is maximum. Consider a point P at a distance x from O . We shall now consider the conditions for a bright or dark fringe at this point.

From right angle triangle S_1QP

$$(S_1P)^2 = (S_1Q)^2 + (QP)^2$$

or $(S_1P)^2 = D^2 + (x-d)^2$ [$\because QP = (x-d)$]

Similarly in right angle triangle S_2RP

$$(S_2P)^2 = (S_2R)^2 + (RP)^2$$

or $(S_2P)^2 = D^2 + (x+d)^2$ [$\because RP = (x+d)$]

$$\therefore (S_2P)^2 - (S_1P)^2 = (x+d)^2 - (x-d)^2 = 4xd$$

or $(S_2P - S_1P)(S_2P + S_1P) = 4xd$

In Young's experiment, D is some thousand times greater than $2d$ or x , so that if $(S_2P + S_1P)$ is replaced by $2D$, the error is not more than a fraction of one percent. Hence

$$(S_2P - S_1P) \cdot 2D = 4xd$$

or $(S_2P - S_1P) = \frac{4xd}{2D} = \frac{2xd}{D}$... (1)

Position and spacing of fringes. Now we shall consider the following two cases :

(i) **Bright fringes.** The point P is bright when the path difference* is a whole number multiple of wavelength λ , i.e.

$$S_2P - S_1P = n\lambda \quad \text{where } n = 0, 1, 2, \dots$$

Substituting the value of $(S_2P - S_1P)$ from equation (1), we have

$$\frac{2xd}{D} = n\lambda \quad \text{or} \quad x = \frac{n\lambda D}{2d} \quad \dots (2)$$

Equation (2) gives the distance of the bright fringes from point O . At O , the path difference is zero,

*We know that when the path difference between the two waves is λ , the phase difference is 2π .

For a path difference of x , the phase difference = $\frac{2\pi x}{\lambda}$.

$\therefore$ Phase differences $\delta = \frac{2\pi x}{\lambda} = \frac{2\pi}{\lambda} \times (\text{path difference})$.

hence there is a bright fringe. The next bright fringes are formed when $n = 1, 2, 3, \dots$ and so on, i.e.,

$$\begin{aligned} \text{When } n = 1 \quad x_1 &= \frac{\lambda D}{2d} \\ n = 2 \quad x_2 &= \frac{2\lambda D}{2d} \\ n = 3 \quad x_3 &= \frac{3\lambda D}{2d} \\ &\dots\dots\dots \\ n = n \quad x_n &= \frac{n\lambda D}{2d} \end{aligned}$$

The distance between any two consecutive bright fringes is

$$x_2 - x_1 = \frac{2\lambda D}{2d} - \frac{\lambda D}{2d} = \frac{\lambda D}{2d} \quad \dots(3)$$

(ii) **Dark fringes.** The point P is dark when the path difference is an odd number multiple of half wavelength, i.e.

$$(S_2P - S_1P) = (2n + 1)\lambda/2$$

where $n = 0, 1, 2, 3, \dots$

$$\text{or } \frac{2x}{D} = \frac{(2n+1)\lambda}{2} \quad \text{or } x = \frac{(2n+1)\lambda D}{4d} \quad \dots(4)$$

Equation (4) gives the distances of the dark fringes from point O. The dark fringes are formed as follows :

$$\begin{aligned} \text{When } n = 0 \quad x_0 &= \frac{\lambda D}{4d} \\ n = 1 \quad x_1 &= \frac{3\lambda D}{4d} \\ n = 2 \quad x_2 &= \frac{5\lambda D}{4d} \\ &\dots\dots\dots \\ n = n \quad x_n &= \frac{(2n+1)\lambda D}{4d} \end{aligned}$$

The distance between any two consecutive dark fringes is

$$x_2 - x_1 = \frac{5\lambda D}{4d} - \frac{3\lambda D}{4d} = \frac{2\lambda D}{4d} = \frac{\lambda D}{2d} \quad \dots(5)$$

Hence the spacing between any two consecutive maxima or minima is the same. This is expressed by

$\beta \left(\beta = \frac{\lambda D}{2d} \right)$ and is known as fringe width. It is ob-

vious that the spacing is directly proportional to D and λ and inversely proportional to $2d$.

26.6. CONDITIONS FOR INTERFERENCE OF LIGHT

To obtain a permanent or stationary interference pattern, the conditions are classified into the following three parts :

- (1) Conditions for sustained interference.
- (2) Conditions for observation of the fringes.
- (3) Conditions for good contrast between maxima and minima.

(1) Condition for sustained interference

(i) *The two sources should be coherent,* i.e., they should vibrate in the same phase or there should be a constant phase difference between them.*

In order to obtain stationary interference pattern, the resultant intensity at any point should be constant with time. The resultant intensity at any point is given by the expression

$$I = a_1^2 + a_2^2 + 2a_1a_2 \cos \delta$$

It is clear from this expression that I will be constant provided δ remains constant. The total phase difference δ consists of two factors δ_1 and δ_2 ($\delta = \delta_1 + \delta_2$) where δ_1 is the initial phase difference between the two sources and δ_2 is the phase difference at that point due to path difference between the two waves. As path difference for a particular point remains constant hence δ_2 remains constant. Thus δ will remain constant if δ_1 remains constant. The explanation also proves that why the two independent sources cannot produce sustained interference. It is the fundamental property of the atoms of the source of light that they are continually undergoing transitions which produce frequency haphazard phase changes in the light waves which they emit. Due to this fact the phase difference between two sources does not remain constant.

(ii) *The two sources must emit continuous waves of the same wavelength and time period.*

If the sources do not emit light of the same period, the intensity at any point will be alternately

* If the phase difference between the two waves emitted from two sources is zero or has a constant value with time, then the two sources are known as coherent sources. However, if the phase difference changes with time, the two sources are known as incoherent sources.

maximum and minimum and hence the interference will not be sustained.

(2) Conditions for observation

(i) The separation between the two sources ($2d$) should be small.

When the separation between the two sources is small, the fringe width ($D\lambda/2d$) is large and the fringes are separately visible. On the other hand if $2d$ is large, fringe width will be small and due to the limited resolving power of eye, the fringes will not be separately visible.

(ii) The distance D between the two sources and screen should be large.

When D is large, fringe width is large and hence they are separately visible.

(iii) The background should be dark.

(3) Conditions for good contrast

(i) The amplitudes of the interfering waves should be equal or nearly equal.

In case of two interfering waves of amplitudes a_1 and a_2 the intensities of maxima and minima are $(a_1 + a_2)^2$ and $(a_1 - a_2)^2$ respectively. If the two amplitudes are equal, the minimum intensity would be zero and there will be a good contrast between maxima and minima. If there is a large difference between a_1 and a_2 , the intensity of minima will not much differ from that of maxima and the contrast will be poor.

(ii) The sources must be narrow i.e., they must be extremely small.

If the sources are not narrow, the interference between waves from different parts of the same source will take place and contrast will be poor.

(ii) The sources should be monochromatic.

If instead of a monochromatic source (single wavelength) a white light source (different wavelengths) is used, then each separate colour (wavelength) will produce its own interference fringes with its own spacing. The different set of fringes due to different colours will overlap on each other and the net effect would be white light. Only a number of limited fringes can be observed in this case. Hence the sources should be monochromatic.

NUMERICAL EXAMPLES

EXAMPLE 1 Two coherent sources of intensity ratio β interfere. Prove that in the interference pattern.

$$\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{\beta}}{1 + \beta}$$

Solution : Let I_1 and I_2 be the intensities and a_1 and a_2 the amplitudes of the disturbances from the two coherent sources. We know that the intensity is proportional to the square of amplitudes, we have

$$\frac{I_1}{I_2} = \frac{a_1^2}{a_2^2} \text{ or } \frac{a_1}{a_2} = \sqrt{\frac{I_1}{I_2}} = \sqrt{\beta} \quad (\text{given})$$

Further $I_{\max} = (a_1 + a_2)^2$ and $I_{\min} = (a_1 - a_2)^2$

$$\begin{aligned} \therefore \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} &= \frac{(a_1 + a_2)^2 - (a_1 - a_2)^2}{(a_1 + a_2)^2 + (a_1 - a_2)^2} \\ &= \frac{2a_1a_2}{a_1^2 + a_2^2} = \frac{2(a_1/a_2)}{(a_1/a_2)^2 + 1} = \frac{2\sqrt{\beta}}{1 + \beta} \end{aligned}$$

EXAMPLE 2 Two coherent sources whose intensity ratio is 81 : 1 produce interference fringes. Deduce the ratio of maximum intensity to minimum intensity.

Solution : Given that

$$\frac{I_1}{I_2} = \frac{a_1^2}{a_2^2} = \frac{81}{1} \text{ or } \frac{a_1}{a_2} = \frac{9}{1} \therefore a_1 = 9a_2$$

$$\begin{aligned} \text{Now, } \frac{I_{\max}}{I_{\min}} &= \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \frac{(9a_2 + a_2)^2}{(9a_2 - a_2)^2} \\ &= \frac{(10a_2)^2}{(8a_2)^2} = \frac{100}{64} = \frac{25}{16} \end{aligned}$$

$$\therefore I_{\max} : I_{\min} = 25 : 16$$

EXAMPLE 3 Find the ratio of intensity at the centre of a bright fringe to the intensity at a point one quarter of the distance between two fringes from the centre.

Solution : We know that

$$\begin{aligned} I &= a_1^2 + a_2^2 + 2a_1a_2 \cos \delta \\ &= a^2 + a^2 + 2a^2 \cos \delta \\ &= 2a^2 (1 + \cos \delta) \quad (a_1 = a_2 = a) \end{aligned}$$

At centre $\delta = 0$, so $I_0 = 2a^2 (1 + 1) = 4a^2$

The phase difference between two consecutive fringes is 2π , therefore, phase difference at a distance one quarter between two fringes will be $2\pi/4 = \pi/2$.

If I_1 is the intensity at a point one quarter of the distance between two fringes from centre, then

$$I_1 = 2a^2 (1 + \cos \pi/2) = 2a^2$$

$$\text{Now } I_0/I_1 = 4a^2/2a^2 = 2$$

EXAMPLE 4 One of the slits in a Young's experiment is wider than the other, so that amplitudes of light reaching the central point of the screen from one slit, acting alone, is twice that from the other slit, acting alone. Find the expression for resultant intensity I_0 at direction θ on the screen.

Solution : We know that

$$I = a_1^2 + a_2^2 + 2a_1a_2 \cos \delta$$

At central point $\delta = 0$, if I_0 be the intensity, then

$$I_0 = a_1^2 + a_2^2 + 2a_1a_2 = (a_1 + a_2)^2 = 9a^2$$

$$(\because a_1 = a \text{ and } a_2 = 2a)$$

$$\therefore a^2 = (I_0/9)$$

For any other direction

$$\text{Path difference} = \frac{2x_n d}{D} = (2d) \sin \theta$$

$$\begin{aligned} \therefore I_\theta &= a^2 + 4a^2 + 4a \cos \left\{ \frac{2\pi}{\lambda} \times (2d \sin \theta) \right\} \\ &= a^2 \left[5 + 4 \cos \left\{ \frac{2\pi}{\lambda} \times 2d \sin \theta \right\} \right] \\ &= \frac{I_0}{9} \left[5 + 8 \cos^2 \left\{ \frac{\pi}{\lambda} \times 2d \sin \theta \right\} - 4 \right] \\ &= \frac{I_0}{9} \left[1 + 8 \cos^2 \left\{ \frac{\pi}{\lambda} 2d \sin \theta \right\} \right] \end{aligned}$$

26.7. TYPES OF INTERFERENCE

The phenomenon of interference is divided into two classes, namely; (1) division of wavefront and (2) division of amplitude.

(1) **Division of wavefront.** The incident wavefront is divided into two parts by utilising the phenomena of reflection, refraction or diffraction. These two parts of the same wavefront travel unequal distances and reunite at some angle to produce interference bands. The Fresnel Biprism, Lloyd's mirror etc. are the examples of this class.

(2) **Division of amplitude.** The amplitude of the incoming beam is divided into two parts either by parallel reflection or refraction. These divided parts reunite after traversing different paths and produce interference. In this case it is not essential to employ a point or a narrow line source but a broad light source may be employed to produce brighter

bands. Newton's rings, Michelson's interferometer etc. come under this class.

26.8. FRESNEL'S BIPRISM

Fresnel used a biprism to show the phenomenon of interference. A biprism is essentially a combination of two acute prisms placed base to base. In fact this combination is obtained from an optically plane glass plate by proper grinding and polishing. The obtuse angle of the prism is about 179° and the other angles are about $30'$ each. The action of the biprism is to produce two coherent images of a given slit which are separated at a certain distance and behave as two coherent sources. To show the phenomenon of interference, a horizontal section of the apparatus is shown in Fig. 26.5.

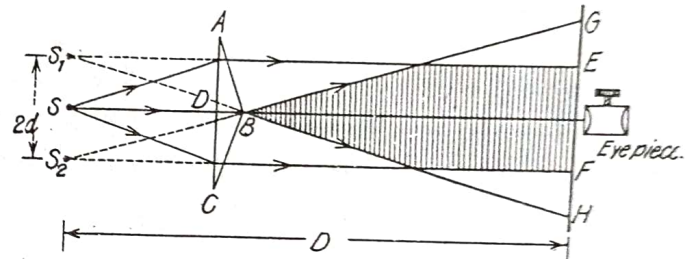


Fig. 26.5

The experimental arrangement consists of an optical bench carrying three stands, the first for an adjustable slit, second for biprism, and the third for a micrometer eyepiece. These stands are adjusted at the same height. The stands can be moved along as well as right angle to the length of optical bench. The slit and biprism can be rotated in their own planes. The slit S is illuminated by a monochromatic source of light such as sodium lamp. The light emerging from the slit falls on the biprism. The edge B of the biprism divides the incident wave-front into two parts. Firstly the one which is passing through the upper half ABD of the biprism is deviated through a small angle towards the lower half of the diagram and appears to diverge from virtual image S₁. Secondly the one which is passing through the lower half CBD is deviated towards the upper half and appears to diverge from the virtual image S₂. The two emergent wavefronts are derived from the same wavefront, hence the fundamental condition of interference is satisfied. S₁ and S₂ serve as coherent sources. The interference fringes are obtained in the overlapping region EF and can be seen by eyepiece.

Determination of wavelength of light

Adjustments. To obtain the interference fringes the following adjustments are made :

(i) The optical bench is levelled with the help of a spirit level.

(ii) Slit, biprism and eyepiece are adjusted to the same height.

(iii) Eyepiece is focussed on its cross-wires.

(iv) The slit is made narrow and vertical, and is illuminated by monochromatic light.

(v) The biprism is moved right angles to the length of optical bench till overlapping region (a patch of bright light) is brought in field of view. The tangent screw of slit is rotated so that the fringes appear.

(vi) The fringes are made clear by rotating the biprism in its own plane with the help of tangent screw provided with the biprism stand.

(vii) The eyepiece is moved away from the biprism, the fringes often shift side-ways. This is known as *lateral shift*. This lateral shift is removed. When the lateral shift is removed, the interference pattern becomes stationary for any position of eyepiece. The reason for lateral shift is that the line joining the slit and biprism is not parallel to the length of optical bench.

Theory. The wavelength λ of the light can be determined by the following formula.

$$\lambda = \frac{\beta 2d}{D},$$

where β = fringe width i.e., spacing between successive maxima or minima.

$2d$ = distance between the two virtual sources.

D = distance between the slit and eyepiece where the fringes are observed.

For the derivation of the formula see article 26.5.

Procedure. In order to determine the wavelength of light, the following measurements are made:

(1) *Measurement of fringe width β .* The vertical cross-wire is adjusted on any bright fringe and the reading of micrometer screw is noted. Now we set the cross-wire one by one on different bright fringes and note down the micrometer screw reading in each case. The mean distance between the two bright fringes β is thus found out.

(2) *Measurement of D .* Distance between slit and eyepiece can be read directly on the optical bench.

This distance is then corrected from bench error which gives D .

(3) *Measurement of $2d$.* To determine $2d$, a lens with focal length less than one-fourth of the distance between biprism and eyepiece is mounted between biprism and eyepiece as shown in Fig. 26.6. The lens is adjusted in two positions L_1 and

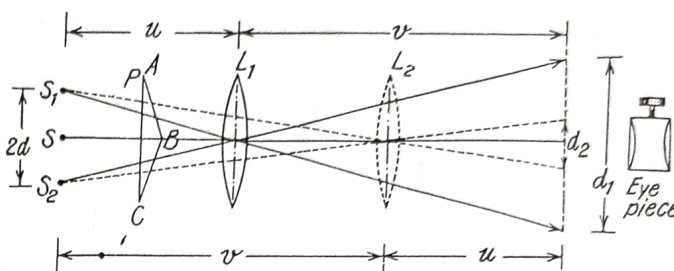


Fig. 26.6

L_2 till sharp images of S_1 and S_2 are obtained in the field of view of the eyepiece. The distances d_1 and d_2 between the real images in two cases are measured. Now $2d$ is calculated as follows:

$$\text{From the figure } \frac{d_1}{2d} = \frac{v}{u} \text{ and } \frac{d_2}{2d} = \frac{u}{v} \\ \therefore 2d = \sqrt{d_1 d_2}$$

Thus wavelength λ can be determined by measuring the values β , D and $2d$.

26.9. WHITE LIGHT FRINGES AND LOCATION OF ZERO-ORDER FRINGE IN BIPRISM EXPERIMENT

White light fringes. We know that white light is a mixture of so many wavelengths varying from 4400 A.U. (violet) to 7500 A.U. (red.) So the coherent sources of monochromatic light are now equivalent to an infinite number of monochromatic sources of different wavelengths.

According to the formula $\beta = \frac{\lambda D}{2d}$, each pair of monochromatic sources of white light produces its own system of fringes with different fringe widths. The resultant interference pattern (Fig. 26.7) consists of:

- Zero order band which is white.
- On both sides of zero order band, few coloured fringes are observed. In coloured fringes, the inner ends are reddish while outer edges are violet.
- After coloured fringes, there is uniform illumination on both sides.

Location of zero order fringe. When a monochromatic source of light is used all the bright fringes are of same colour. Hence it is impossible to locate the zero order fringe. In order to locate the zero order fringe, first of all white light fringes are

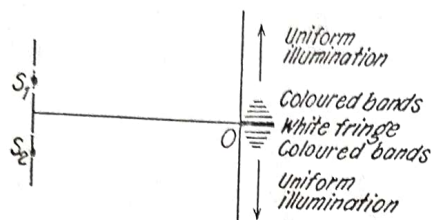


Fig. 26.7

obtained and the cross-wire is fixed on the zero order fringe (which is white in this case while others are coloured). Now white light source is replaced by the monochromatic source. The fringe now coinciding with cross-wire will be zero order fringe (central fringe) in monochromatic light system.

26.10. DETERMINATION OF THE THICKNESS OF A THIN SHEET OF TRANSPARENT MATERIAL

Displacement of fringes. The biprism experiment can be used to determine the thickness of a thin sheet of transparent material such as glass, mica, etc. When a thin transparent sheet (say Mica) of thickness t and refractive index μ is introduced in the path of one of the interfering beams (Fig. 26.8), the entire fringe system is displaced through a con-

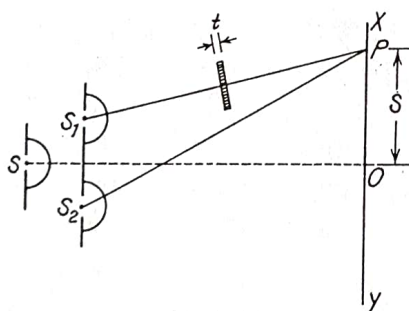


Fig. 26.8

stant distance towards the path of the beam in which the plate is introduced. It should be noted here that the fringe width remains unchanged. Now we consider the path S_1P . The length $(S_1P - t)$ of this path is travelled in air with the velocity of light c while the length t is travelled in mica with velocity c_g , where c_g is the velocity of light in mica. The time taken by the light to cover this distance is given by $[(S_1P - t)/c] + [t/c_g]$. We

also know that $c/c_g = \mu$ or $c_g = c/\mu$. Hence time taken by S_1 to P .

$$= \frac{S_1P - t}{c} + \frac{t\mu}{c} = \frac{S_1P + (\mu - 1)t}{c}$$

Thus the path S_1 to P is equivalent to an air path $S_1P + (\mu - 1)t$

Now the path difference at P

$$= \text{path } S_2P \text{ in air} - \text{path } S_1 \text{ to } P$$

$$= S_2P - [S_1P + (\mu - 1)t]$$

$$= S_2P - S_1P - (\mu - 1)t = \frac{2xd}{D} - (\mu - 1)t \quad \dots(1)$$

$$\left[\because S_2P - S_1P = \frac{2xd}{D} \right]$$

$$\text{For } n\text{th maxima, path difference} = n\lambda \quad \dots(2)$$

From equations (1) and (2), for n th maxima

$$\frac{2x_n d}{D} - (\mu - 1)t = n\lambda$$

$$\text{or } x_n = \frac{D}{2d} [n\lambda + (\mu - 1)t] \quad \dots(3)$$

where x_n denotes the position of n th maxima.

In the absence of the plate ($t = 0$), the n th maxima is $= \frac{Dn\lambda}{2d}$.

If S denotes the displacement of n th maxima by introducing the mica sheet, then

$$S = \frac{D}{2d} [n\lambda + (\mu - 1)t] - \frac{Dn\lambda}{2d}$$

$$\text{or } S = \frac{D}{2d} (\mu - 1)t \quad \dots(4)$$

Equation (4) is independent of n , hence the displacement of each maxima is the same.

Thickness of mica sheet. The displacement S of any maxima by introducing a mica sheet of thickness t is given by

$$S = \frac{D}{2d} (\mu - 1)t$$

$$\text{or } t = \frac{S \times 2d}{D (\mu - 1)} \quad \dots(5)$$

Here $2d$ and D can be measured by biprism experiment. To determine S , the monochromatic source of light is replaced by white light source and the crosswire is set on the white fringe. The reading of micrometer screw is noted. Now the mica sheet is introduced in one of the two interfering beams. On

doing so, the white fringe shifts from its previous position. The crosswire is again set on white fringe and micrometer reading is noted. The difference of the two readings gives the displacement S . Substituting these values in equation (5), t can be calculated.

NUMERICAL EXAMPLES

EXAMPLE 5 Two straight and narrow parallel slits 1 mm apart are illuminated by monochromatic light. Fringes formed on the screen held at a distance of 100 cm from the slits are 0.50 mm apart. What is the wavelength of light?

Solution : According to the problem $\beta = 0.50$ mm. = 0.05 cm, $2d = 1$ mm = 0.1 cm, and $D = 100$ cm.

The fringe width is given by

$$\beta = \frac{\lambda D}{2d} \text{ or } \lambda = \frac{\beta \times 2d}{D}$$

Substituting the given values, we have

$$\lambda = \frac{0.005 \times 0.1}{100} = 5 \times 10^{-5} \text{ cm} \\ = 5000 \text{ \AA}.$$

EXAMPLE 6 Two coherent sources are placed 0.9 mm apart and the fringes are observed one metre away. Calculate the wavelength of the monochromatic light used if it produces the second dark fringe at a distance of 10 mm from the central fringe.

Solution : Here $2d = 0.9$ mm = 0.09 cm, $D = 1$ metre = 100 cm, $n = 2$, $x_2 = 10$ mm = 1 cm.

For a dark fringe.

$$x_n = \frac{D}{2d} (2n+1) \frac{\lambda}{2} \text{ or } \lambda = \frac{4d}{D} \frac{x_n}{(2n+1)}$$

Substituting the given values, we have

$$\lambda = \frac{0.09 \times 2 \times 1}{100 \times 5} = 3.6 \times 10^{-5} \text{ cm}.$$

EXAMPLE 7 A biprism is placed 5 cm from a slit illuminated by sodium light ($\lambda = 5890 \text{ \AA}$). The width of the fringes obtained on a screen 75 cm from the biprism is 9.424×10^{-2} cm. What is the distance between the two coherent sources?

Solution : Here $\lambda = 5890 \text{ \AA} = 5890 \times 10^{-8}$ cm, $\beta = 9.424 \times 10^{-2}$ cm, $D = 5 + 75 = 80$ cm, $2d = ?$

$$\text{We know that } \beta = \frac{\lambda D}{2d}$$

$$\therefore 2d = \frac{\lambda D}{\beta} = \frac{5890 \times 10^{-8} \times 80}{9.424 \times 10^{-2}} = 0.05 \text{ cm}$$

EXAMPLE 8 The distances between the slit and biprism and between the biprism and the screen are 50 cm each. The angle of biprism is 179° and its refractive index is 1.5. If the distance between successive fringes is 0.0135 cm, calculate the wavelengths of light.

Solution : Here $\beta = 0.0135$ cm, $D = 50 + 50 = 100$ cm, $\mu = 1.5$.

We know that $2d = 2a(\mu - 1)A$, where

$$A = \frac{180^\circ - 179^\circ}{2} = \left(\frac{1}{2}\right)^\circ = \frac{1}{2} \times \frac{\pi}{180} = \frac{\pi}{360} \text{ radians}$$

and $a = 50$

$$\therefore 2d = 2 \times 50 \times (1.5 - 1) \times \frac{\pi}{360} \text{ cm}.$$

$$\text{Now } \lambda = \frac{\beta 2d}{D} = \frac{0.0135 \times 2 \times 50 \times (1.5 - 1)\pi}{100 \times 360}$$

$$= 5893 \times 10^{-8} \text{ cm} = 5893 \text{ \AA}.$$

EXAMPLE 9 Fresnel's biprism having an angle of 1° and index of refraction 1.50 forms interference fringes on a screen placed 80 cms from the biprism. If the distance between the slit and the biprism is 30 cms find the fringe separation, when wavelength of light used is 5000 A.U. Calculate the fringe separation when the distance of 80 cms and 30 cms are interchanged.

Solution : Here $A = 1^\circ = \frac{\pi}{180^\circ}$ radians

$$[\because 180^\circ = \pi \text{ radians}]$$

$$\mu = 1.5, \lambda = 5000 \text{ A.U.} = 5 \times 10^{-5} \text{ cm}$$

(Distance between slit and biprism).

$$a = 30 \text{ cms}$$

(Distance between biprism and screen),

$$b = 80 \text{ cms. } \therefore D = 30 + 80 = 110 \text{ cms.}$$

$$\text{Now } \beta = \frac{\lambda D}{2d} = \frac{\lambda D}{2a(\mu - 1)A}$$

$$= \frac{5 \times 10^{-5} \times 110 \times 180}{2 \times 30 \times (1.5 - 1) \times \pi} = 0.0105 \text{ cm}.$$

In second case $a = 80$, $b = 30$, i.e., $D = 110$

$$\therefore \beta = \frac{5 \times 10^{-5} \times 110 \times 180}{3 \times 80 \times (1.5 - 1) \times \pi} = 0.00394 \text{ cm}$$

INTERFERENCE

26.11

mm from the second dark fringe from the centre pattern. Find the wavelength of light used.

Solution : The distance of n th bright fringe from the central fringe is given by

$$x_n = n \frac{\lambda D}{2d} = n\beta$$

where $\beta = (\lambda D / 2d)$ is the fringe width

For 9th bright fringe, $x_9 = 9\beta$

The distance of the n th dark fringe from the central fringe is given by

$$x'_n = n \frac{(2n-1)\lambda D}{4d} = \left(n - \frac{1}{2}\right) \frac{\lambda D}{2d} = \left(n - \frac{1}{2}\right)\beta$$

For 2nd dark fringe, $x'_2 = \left(\frac{3}{2}\right)\beta$

From eqs. (1) and (2), we get

$$x_9 - x'_2 = 9\beta - \frac{3}{2}\beta = \frac{15}{2}\beta$$

$$\text{or } 8.835 \text{ mm} = \frac{15}{2}\beta \quad (\because x_9 - x'_2 = 8.835 \text{ mm})$$

Solving we get $\beta = 1.178 \text{ mm} = 1.178 \times 10^{-3} \text{ m}$

$$\text{Now } \lambda = \beta \frac{2d}{D} = (1.178 \times 10^{-3}) \times \left(\frac{0.5 \times 10^3}{1}\right) = 5.89 \times 10^{-7} \text{ m} = 5890 \text{ \AA}$$

EXAMPLE 15 A beam of light consisting of two wavelengths $6500 \text{ \AA}$ and $5200 \text{ \AA}$ is used to obtain interference fringes in a Young's double slit experiment :

(i) Find the distance of the third bright fringe on the screen from the central maximum for the wave length $6500 \text{ \AA}$.

(ii) What is the least distance from the central maximum where the bright fringes due to both the wavelengths coincide ?

The distance between the slits is 2 mm and distance between the plane of the slits and the screen is 120 cm .

Solution : (i) For bright fringes

$$x_n = n \frac{\lambda D}{2d}$$

Here $n = 3$, $\lambda = 6500 \times 10^{-8} \text{ cm}$, $D = 120 \text{ cm}$ and $2d = 2 \text{ mm} = 0.2 \text{ cm}$

$$\therefore x_3 = 3 \times \frac{(6500 \times 10^{-8}) (120)}{0.2} = 1.17 \text{ mm}$$

(ii) The bright fringes of both the wavelengths will coincide when for a certain distance, n bright fringes due to one wavelength is equal to $(n+1)$

bright fringes due to other wavelength. If β is the fringe width, then

$$n \frac{\lambda_1 D}{2d} = (n+1) \frac{\lambda_2 D}{2d} \quad \text{or } n\lambda_1 = (n+1)\lambda_2$$

$$n(6500 \times 10^{-8}) = (n+1)(5200 \times 10^{-8})$$

$$n = 4$$

$$\text{Distance } x_4 = 4 \cdot \frac{\lambda_1 D}{2d} = \frac{4 \times (6500 \times 10^{-8}) 120}{0.2} = 1.56 \text{ mm}$$

EXAMPLE 16 A thin sheet of glass ($\mu = 1.5$) of 6 microns thickness introduced in the path of interfering beams in a biprism arrangement shifts the central fringe to a position normally occupied by the fifth. Find the wavelength of light used. (1 micron = 10^{-4} cm).

Solution : Here $\mu = 1.5$, $t = 6 \text{ microns} = 6 \times 10^{-4} \text{ cm}$, $S = 5\beta$.

The thickness t of the glass plate is given by

$$t = \frac{S\lambda}{(\mu-1)\beta} \quad \text{or } \lambda = \frac{t(\mu-1)\beta}{S}$$

$$\begin{aligned} \therefore \lambda &= \frac{6 \times 10^{-4} \times (1.5-1)\beta}{5\beta} \\ &= 6 \times 10^{-5} = 6000 \times 10^{-8} \text{ cm} \\ &= 6000 \text{ \AA} \end{aligned}$$

EXAMPLE 17 The central fringe of the interference produced by light of wavelength $6000 \text{ \AA}$ is shifted to the position of 5th bright fringe by introducing a thin glass plate of refractive index 1.5. Calculate the thickness of the plate.

Solution : We know that the shift S of the central or of any other fringe is given by

$$S = \frac{D}{2d} (\mu-1)t = \frac{\beta}{\lambda} (\mu-1)t$$

where β is the fringe width.

According to the given problem.

$$S = 5\beta, \lambda = 6000 \text{ \AA} = 6 \times 10^{-5} \text{ cm} \text{ and } \mu = 1.5.$$

$$\therefore 5\beta = \frac{\beta(1.5-1)t}{6 \times 10^{-5}}$$

$$\text{or } t = \frac{5 \times 6 \times 10^{-5}}{0.5} = 6 \times 10^{-4} \text{ cm.}$$

EXAMPLE 18 In a double slit arrangement, fringes are produced using light of wavelength $4800 \text{ \AA}$. One slit is covered by a thin plate of glass of refractive index 1.4

$$n \sin \alpha = \lambda$$

$$1 - x_n$$

$$\frac{\lambda}{2\alpha} = \frac{\lambda}{2\alpha} \quad \dots(5)$$

measured in radians.

AN EXTENDED SOURCE

film which is illuminated
(26.16).

26.17 NEWTON'S RINGS

When a plano-convex lens with its convex surface is placed on a plane glass plate, an air film of gradually increasing thickness is formed between the two. The thickness of the film at the point of contact is zero. If monochromatic light is allowed to fall normally, and the film is viewed in reflected light, alternate dark and bright rings concentric around the point of contact between the lens and glass plate are seen. Since the phenomenon was

first described by Newton that is why the rings are known as Newton's rings after his name.

Experimental arrangement. The experimental arrangement of obtaining Newton's rings is shown in Fig. 26.18. L is a planoconvex lens of radius of

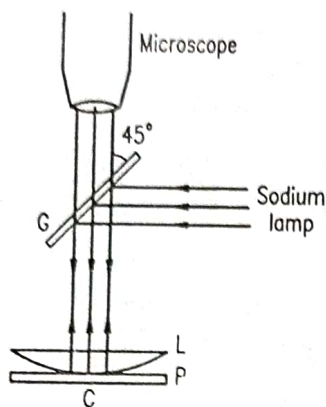


Fig. 26.18

large curvature. This lens with its convex surface is placed on a plane glass plate P . The lens makes contact with the plate at C . Light from an extended monochromatic source such as sodium lamp falls on a glass plate G held at an angle 45° with the vertical. The glass plate G reflects normally a part of the incident light towards the air film enclosed by the lens L and the glass plate P . A part of the incident light is reflected by the curved surface of the lens L and a part is transmitted which is reflected back from the plane surface of the plate. These two reflected rays interfere and give rise to an interference pattern in the form of circular rings. These rings are localised in the air film, and can be seen with a microscope focussed on the film.

Explanation of the formation of Newton's rings. Newton's rings are formed due to interference between the waves reflected from the top and bottom surfaces of the air film formed between the plates. The formation of Newton's rings can be explained with the help of Fig. 26.19. AB is a

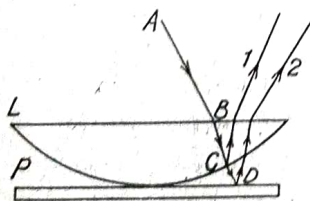


Fig. 26.19

*Actual path difference = $2\mu t \cos(r + \alpha) + \lambda/2$. Due to large radius of curvature of plano-convex lens, α is extremely small and hence neglected here.

monochromatic ray of light which falls on the system. A part is reflected at C (glass-air boundary) which goes out in the form of ray 1 without any phase reversal. The other part is refracted along CD . At point D it is again reflected and goes out in the form of ray 2 with a phase reversal of π . The reflected rays 1 and 2 are in a position to produce interference fringes as they have been derived from the same ray AB and hence fulfil the condition of interference. As the rings are observed in the reflected light, the path difference* between them is $(2\mu t \cos r + \lambda/2)$. For air film $\mu = 1$ and for normal incidence $r = 0$, hence in the case, the path difference is $(2t + \lambda/2)$. At the point of contact $t = 0$, and the path difference is $\lambda/2$, which is the condition of minimum intensity. Thus the central spot is dark. For n th maximum, we have

$$2t + \lambda/2 = n\lambda$$

This expression shows that a maximum of a particular order n will occur for a constant value of t . In case of this system, t remains constant along a circle. Thus the maximum is in the form of a circle. For different values of t , different maxima will occur. In a similar way, this can be shown that minima are also in the circular form.

Theory : (1) Newton's rings by reflected light. Now we shall calculate the diameters of dark and bright rings. Let LOL' be the lens placed on a glass plate AB . The curved surface LOL' is the part of spherical surface (shown dotted in Fig. 26.20) with

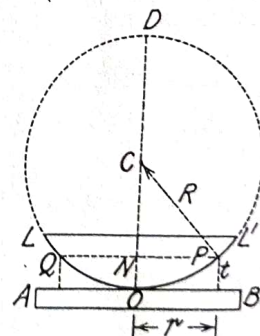


Fig. 26.20

centre at C . Let R be the radius of curvature and r be the radius of Newton's ring-corresponding to the constant film thickness t . As discussed above,

$$2t + \lambda/2 = n\lambda$$

$$\text{or } 2t = (2n - 1) \lambda/2 \text{ for the bright ring}$$

where $n = 1, 2, 3, \dots$ etc.

and $2t = n\lambda$ for dark ring

where $n = 0, 1, 2, 3 \dots$ etc.

From the property of the circle

$$NP \times NQ = NO \times ND$$

Substituting the values

$$r \times r = t \times (2R - t) = 2Rt - t^2 \approx 2Rt \quad (\text{approximately})$$

$$\therefore r^2 = 2Rt \text{ or } t = r^2/2R.$$

Thus for a bright ring

$$2 \frac{r^2}{2R} = (2n-1) \frac{\lambda}{2}$$

$$\text{or } r^2 = \frac{(2n-1)\lambda R}{2}$$

Replacing r by $D/2$, we get the diameter of n th bright ring as

$$\frac{D^2}{4} = \frac{(2n-1)\lambda R}{2}$$

$$\text{or } D = \sqrt{(2\lambda R)} \sqrt{(2n-1)}$$

$$\text{or } D \approx \sqrt{(2n-1)}$$

Thus the diameters of the bright rings are proportional to the square roots of odd natural numbers as $(2n-1)$ is an odd number.

Similarly for a dark ring

$$2 \frac{r^2}{2R} = n\lambda$$

$$\text{or } r^2 = n\lambda R$$

$$\text{or } D^2 = 4n\lambda R$$

$$\text{or } D = 2\sqrt{n\lambda R} \propto \sqrt{n}$$

Thus diameters of dark rings are proportional to the square roots of natural numbers.

It can be shown that fringe width decreases with the order of the fringe and fringes get closer with increase in their order.

The diameters of 16th and 9th dark rings are given as

$$D_{16} = 2\sqrt{16\lambda R} = 8\sqrt{\lambda R}$$

$$D_9 = 2\sqrt{9\lambda R} = 6\sqrt{\lambda R}$$

$$\therefore D_{16} - D_9 = 8\sqrt{\lambda R} - 6\sqrt{\lambda R} = 2\sqrt{\lambda R}$$

This is shown in Fig. 26.21.

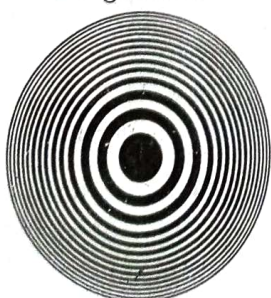


Fig. 26.21

(ii) **Newton's rings by transmitted light.** In case of transmitted light

$$2t = n\lambda \text{ for bright rings}$$

$$\text{and } 2t = (2n-1)\lambda/2 \text{ for dark rings.}$$

For bright rings

$$\therefore 2 \times \frac{r^2}{2R} = n\lambda \text{ or } r^2 = n\lambda R$$

$$\text{or } D = 2\sqrt{n\lambda R} \propto \sqrt{n}$$

For dark rings

$$\therefore 2 \times \frac{r^2}{2R} = (2n-1) \frac{\lambda}{2} \text{ or } r^2 = \frac{(2n-1)\lambda R}{2}$$

$$\therefore D = \sqrt{2\lambda R} \times \sqrt{(2n-1)} \propto \sqrt{(2n-1)}$$

Thus in case of transmitted light, the central ring is bright (Fig. 26.22). The rings are just opposite to the rings in reflected light.

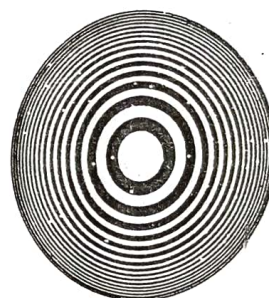


Fig. 26.22

26.18 DETERMINATION OF WAVE-LENGTH OF SODIUM LIGHT USING NEWTON'S RINGS

Experimental arrangement. The experimental arrangement is shown in Fig. 26.18. [see article 26.17, experimental arrangement].

Theory. Let R be the radius of curvature of the surface in contact with the plate, λ the wavelength of light used and D_n and D_{n+p} the diameters of n th and $(n+p)$ th dark rings respectively, then

$$D_n^2 = 4n\lambda R$$

$$\text{and } D_{n+p}^2 = 4(n+p)\lambda R$$

$$\therefore D_{n+p}^2 - D_n^2 = 4p\lambda R$$

$$\text{or } \lambda = \frac{D_{n+p}^2 - D_n^2}{4pR} \quad \dots(1)$$

Using this formula λ can be determined.

Procedure. First of all the eyepiece of the microscope is adjusted on its cross wires. Now the distance of the microscope from the film is adjusted at the rings with dark centre in well focus.

26.20

The centre of the cross-wires is adjusted at the centre of the fringe pattern. By counting the number of fringes, the microscope is moved to the extreme left of the pattern and the cross wire is adjusted tangentially in the middle of a clearly n th (say 20th) bright or dark fringe. The reading of micrometer screw (attached with eyepiece) is noted. The microscope is now moved to the right and the readings of micrometer screw are noted successively at $(n-2)$ th (say 18th), $(n-4)$ th (say 16th) ... rings etc. till we are very near to the central dark spot.

Again crossing the central dark spot in the same direction the readings corresponding to... $(n-4)$ th (16th), $(n-2)$ th (18th) .. n th (20th) rings are noted on other side. Now a graph is plotted between number of rings n and the square of the corresponding diameter. The graph is shown in Fig. 26.23.

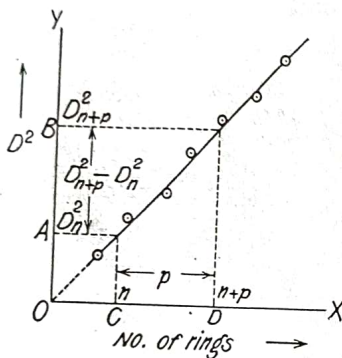


Fig. 26.23

From the graph

$$\frac{D_{n+p}^2 - D_n^2}{p} = \frac{AB}{CD}$$

The radius R of the plano-convex lens can be obtained with the help of spherometer using the following formula

$$R = \frac{l^2}{6h} + \frac{h}{2},$$

where l is the distance between the two legs of the spherometer and h is the difference of the readings of the spherometer when it is placed on the lens as well as when placed on plane surface.

The radius of curvature of the plano-convex lens can also be determined by Boy's method.

Substituting these values in the above derived formula (1), λ can be calculated.

26.19 DETERMINATION OF REFRACTIVE INDEX OF A LIQUID

First of all the experiment is performed when there is an air film between the glass plate and plano-convex lens. The system is placed in a metal container. The diameters of n th and $(n+p)$ th rings are determined with the help of a travelling microscope as discussed in the previous article. So, when there is air film between glass plate and plano-convex lens, we have

$$D_n^2 = 4n\lambda R \text{ and } D_{n+p}^2 = 4(n+p)\lambda R$$

$$\therefore D_{n+p}^2 - D_n^2 = 4p\lambda R \quad \dots(1)$$

Now the liquid whose refractive index is to be determined is poured in the container without disturbing the whole arrangement. Again the diameters of n th ring and $(n+p)$ th ring are determined. So, when there is a liquid film between glass plate and plano-convex lens, we have

$$D'_n{}^2 = \frac{4n\lambda R}{\mu} \text{ and } D'_{n+p}{}^2 = \frac{4(n+p)\lambda R}{\mu}$$

$$\therefore D'_{n+p}{}^2 - D'_n{}^2 = \frac{4p\lambda R}{\mu} \quad \dots(2)$$

From equations (1) and (2)

$$\mu = \frac{D_{n+p}^2 - D_n^2}{D'_{n+p}{}^2 - D'_n{}^2} \quad \dots(3)$$

Using this formula μ can be calculated.

26.20 NEWTON'S RINGS FORMED BY TWO CURVED SURFACES

Case (I). When a plano-convex lens of radius of curvature R_1 is placed on a plano-convex lens of radius of curvature R_2 .

Consider the case of two curved surfaces of radii of curvatures R_1 and R_2 placed in such a way that they are in contact at O as shown in Fig. 26.24. An

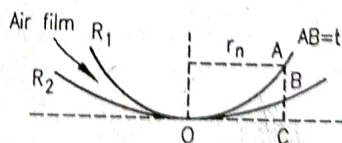


Fig. 26.24

air film is enclosed between the two surfaces. Now a system of dark and bright rings is formed which can be viewed by a travelling microscope. Let t be

NUMERICAL EXAMPLES

EXAMPLE 20 A parallel beam of light ($\lambda = 5890 \times 10^{-8}$ cm) is incident on a thin glass plate ($\mu = 1.5$) such that the angle of refraction into the plate is 60° . Calculate the smallest thickness of the glass plate which will appear dark by reflection.

Solution : Here $\mu = 1.5$, $r = 60^\circ$, $n = 1$ and $\lambda = 5890 \times 10^{-8}$ cm

We know that $2 \mu t \cos r = n \lambda$

$$\begin{aligned} \text{or } t &= \frac{n \lambda}{2 \mu \cos r} = \frac{1 \times 5890 \times 10^{-8}}{2 \times 1.5 \times \cos 60} \\ &= \frac{1 \times 5890 \times 10^{-8}}{2 \times 1.5 \times 0.5} \\ &= 3.926 \times 10^{-5} \text{ cm.} \end{aligned}$$

EXAMPLE 21 A drop of volume 0.2 cc is dropped on a surface of a tank of water of area 1 sq. metre. The film spreads uniformly over the whole surface and white light which is incident normally, is observed through a spectrometer. The spectrum is seen to contain one dark

band whose centre has wavelength 5.5×10^{-5} cm in air. Find the refractive index of oil.

Solution : Since the drop of volume 0.2 cc spreads uniformly over one meter², therefore the thickness of the film so formed is given by

$$t = \frac{0.2}{(100)^2} = 2 \times 10^{-5} \text{ cm.}$$

The film appears dark by reflected light for wavelengths given by

$$2 \mu t \cos r = n \lambda$$

Here for normal incidence $r = 0$, $\cos r = 1$, $n = 1$ and $\lambda = 5.5 \times 10^{-5}$ cm.

$$\mu = \frac{n \lambda}{2 t \cos r} = \frac{1 \times 5.5 \times 10^{-5}}{2 \times 2 \times 10^{-5} \times 1} = \frac{5.5}{4} = 1.375.$$

EXAMPLE 22 While light falls normally on a film of soapy water whose thickness is 5×10^{-5} cm and $\mu = 1.33$. Which wavelength in the visible region will be reflected most strongly?

Solution : The condition of maxima is given by

$$2 \mu \cos r = (2n + 1) \lambda / 2$$

where $n = 0, 1, 2, 3, \dots$

Here $t = 5 \times 10^{-5}$ cm, $\mu = 1.33$, $r = 0^\circ$

i.e., $\cos r = 1$.

$$\text{Now } \lambda = \frac{4 \mu t \cos r}{(2n + 1)} = \frac{4 \times 1.33 \times 5 \times 10^{-5}}{(2n + 1)}$$

By substituting the values of $n = 0, 1, 2, \dots$ we get a series of wavelengths which shall be predominantly reflected by the film.

$$\text{for } n = 0, \lambda_1 = \frac{4 \times 1.33 \times 5 \times 10^{-5}}{1}$$

$$= 26.66 \times 10^{-5} \text{ cm}$$

$$n = 1, \lambda_2 = \frac{4 \times 1.33 \times 5 \times 10^{-5}}{3}$$

$$= 8.866 \times 10^{-5} \text{ cm}$$

$$n = 2, \lambda_3 = \frac{4 \times 1.33 \times 5 \times 10^{-5}}{5}$$

$$= 5.320 \times 10^{-5} \text{ cm}$$

$$n = 3, \lambda_4 = \frac{4 \times 1.33 \times 5 \times 10^{-5}}{7}$$

$$= 3.800 \times 10^{-5} \text{ cm.}$$

Out of these wavelengths only 5.320×10^{-5} cm or 5320×10^{-8} cm lies within the visible region

(4000×10^{-8} cm to 7500×10^{-8} cm). Hence $5230 \text{ \AA}$ is the most strongly reflected wavelength.

EXAMPLE 23 A soap film of refractive index $4/3$ and of thickness 1.5×10^{-4} cm is illuminated by white light incident at an angle of 45° . The light reflected by it is examined by a spectroscope in which is found a dark band corresponding to a wavelength of 5×10^{-5} cm. Calculate the order of interference band.

Solution : In case of dark band $2 \mu t \cos r = n \lambda$

$$\therefore n = \frac{2 \mu t \cos r}{\lambda}$$

Here $\mu = 4/3$, $t = 1.5 \times 10^{-4}$ cm. and $\lambda = 5 \times 10^{-5}$ cm.

$$\text{Further } \sin r = \frac{\sin i}{\mu} = \frac{\sin 45^\circ}{4/3} = \frac{1}{\sqrt{2}} \times \frac{3}{4}$$

$$\therefore \cos r = \sqrt{1 - \sin^2 r} = \sqrt{1 - \left(\frac{9}{32}\right)} = \sqrt{\frac{23}{32}} = \frac{1}{4} \sqrt{\frac{23}{2}}$$

$$\text{Now } n = \frac{2 \times 4 \times 1.5 \times 10^{-4} \times \sqrt{23}}{3 \times 4 \sqrt{2} \times 5 \times 10^{-5}} = 6.782.$$

EXAMPLE 24 White light is incident on a soap film at an angle $\sin^{-1} 4/5$ and the reflected light on examination by a spectroscope shows dark bands. Two consecutive dark bands correspond to wavelengths 6.1×10^{-5} and 6.0×10^{-5} cm. If the refractive index of the film be $4/3$, calculate the thickness.

Solution : For dark bands in reflected light $2 \mu t \cos r = n \lambda$.

When the observed dark bands are consecutive, their orders will differ by one. Let n and $(n + 1)$ be the orders of the dark bands due to wavelengths λ_1 and λ_2 respectively.

Then

$$2 \mu t \cos r = n \lambda_1 = (n + 1) \lambda_2$$

$$\text{or } n = \frac{\lambda_2}{\lambda_1 - \lambda_2}$$

$$\therefore 2 \mu \cos r = n \lambda_1 = \frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2}$$

$$\text{or } t = \frac{\lambda_1 \lambda_2}{2 \mu \cos r (\lambda_1 - \lambda_2)}$$

Here $\lambda_1 = 6.1 \times 10^{-5}$, $\lambda_2 = 6.0 \times 10^{-5}$, $\mu = 4/3$ and

$$\sin i = 4/5$$

$$\begin{aligned}\cos r &= \sqrt{1 - \sin^2 r} = \sqrt{1 - \left(\frac{\sin i}{\mu}\right)^2} \\ &= \frac{\sqrt{(\mu^2 - \sin^2 i)}}{\mu}\end{aligned}$$

$$\begin{aligned}\text{Now } t &= \frac{\lambda_1 \lambda_2 \times \mu}{2 \mu \sqrt{(\mu^2 - \sin^2 i)} \times (\lambda_1 - \lambda_2)} \\ &= \frac{\lambda_1 \lambda_2}{2 \sqrt{(\mu^2 - \sin^2 i)} \times (\lambda_1 - \lambda_2)}\end{aligned}$$

$$\begin{aligned}\text{or } t &= \frac{6.1 \times 10^{-5} \times 6.0 \times 10^{-5}}{2 \times \sqrt{\left(\frac{16}{9} - \frac{16}{25}\right)} \times 0.1 \times 10^{-5}} \\ &= 0.0017 \text{ cm.}\end{aligned}$$

EXAMPLE 25 Fringes of equal thickness are observed in a thin glass wedge of refractive index 1.52. The fringe spacing is 1 mm and wavelength of light is 5893 Å. Calculate the angle of wedge in seconds of an arc.

Solution : Here $\mu = 1.52$, $\lambda = 5893 \times 10^{-8}$ cm, fringe width $\beta = 1 \text{ mm} = 0.1 \text{ cm}$.

$$\text{Now } \beta = \frac{\lambda}{2 \mu \alpha}$$

where α is the angle of wedge.

$$\begin{aligned}\therefore \alpha &= \frac{\lambda}{2 \mu \beta} = \frac{5893 \times 10^{-8}}{2 \times 1.52 \times 0.1} \text{ radian} \\ &= \frac{5893 \times 10^{-8} \times 180 \times 7 \times 60 \times 60}{0.304 \times 22}\end{aligned}$$

seconds of an arc

$$= 39.96 \text{ seconds of an arc.}$$

EXAMPLE 26 If the angle of a wedge is 0.25° and the wavelengths of sodium lines are 5890 Å and 5896 Å, find the distance from the apex at which the maxima due to two wavelengths first coincide when observed in reflected light.

Solution : Let t be the thickness of the wedge at the point where the maximum of 5896 Å coincides with the maximum of 5890 Å. For the maximum intensity in the reflected light, we have

$$2t = (2n + 1) (5896/2) \text{ Å} \quad (\because \mu = 1)$$

$$\text{and } 2t = (2n + 3) (5890/2) \text{ Å}$$

$$\text{Thus } (2n + 1) (5896/2) = (2n + 3) (5890/2)$$

$$\text{Solving, we get } n = 981$$

$$\therefore 2t = (2 \times 981 + 1) (5896/2)$$

$$\begin{aligned}\text{Solving, we get } t &= \frac{1963 \times 5896}{4} \times 10^{-8} \text{ cm} \\ &= 0.029 \text{ cm}\end{aligned}$$

Let x cm be the distance from the apex at which the maxima of two wavelengths coincide. If θ be the wedge angle then

$$t = x \tan \theta = x \theta \text{ (when } \theta \text{ is small)}$$

$$\therefore x = t/\theta$$

$$\text{Here } \theta = 0.25^\circ = \pi/180 \times 0.25 \text{ radian}$$

$$\begin{aligned}\therefore x &= \frac{0.029}{(\pi/180) \times 0.25} \\ &= \frac{0.029 \times 180}{3.14 \times 0.25} = 6.63 \text{ cm}\end{aligned}$$

EXAMPLE 27 A square piece of cellophane film with index of refraction 1.5 has a wedged shaped section so that its thickness at two opposite sides is t_1 and t_2 . If with a light of $\lambda = 6000$ Å, the number of fringes appearing on the film is 10, calculate the difference $(t_2 - t_1)$.

Solution : Here $\mu = 1.5$, $\lambda = 6000$ Å.

If n be the order of the fringes at one end where the thickness is t_1 , then the order of the fringes at the other end where the thickness is t_2 will be $(n + 10)$. Using the formula $2 \mu t \cos r = n \lambda$ at two ends, we have

$$2 \mu t_1 \cos r = n \lambda \quad \dots(1)$$

$$2 \mu t_2 \cos r = (n + 10) \lambda \quad \dots(2)$$

From equations (1) and (2) on subtraction, we get

$$2 \mu (t_2 - t_1) \cos r = 10 \lambda$$

$$\text{or } (t_2 - t_1) = \frac{10 \lambda}{2 \mu \cos r} = \frac{10 \lambda}{2 \mu}$$

$\therefore$ For normal incidence $r = 0$ and $\cos r = 1$

$$\therefore (t_2 - t_1) = \frac{10 \times 6000 \times 10^{-8}}{2 \times 1.5} = 2 \times 10^{-4} \text{ cm.}$$

EXAMPLE 28 In a Newton's ring experiment, the diameter of the 5th ring was 0.3 cm and the diameter of 25th ring was 0.8 cm. If the radius of curvature of the plano convex lens is 100 cm, find the wavelength of light used.

$$\text{Solution : We know that } \lambda = \frac{D_{n+p}^2 - D_n^2}{4 p R}$$

$$\text{Here } D_{25} = 0.8 \text{ cm, } D_5 = 0.3 \text{ cm.}$$

$$p = 25 - 5 = 20, \text{ and } R = 100 \text{ cm}$$

Substituting these values, we have

$$\begin{aligned}\lambda &= \frac{(0.8)^2 - (0.3)^2}{4 \times 20 \times 100} \\ &= 4.87 \times 10^{-5} \text{ cm.}\end{aligned}$$

EXAMPLE 29 A Newton's ring arrangement is used with a source emitting two wavelengths $\lambda_1 = 6 \times 10^{-5}$ cm and 4.5×10^{-5} cm and it is found that n th dark ring due to λ_2 coincides with $(n+1)$ th dark ring for λ_1 . If radius of curvature of the curved surface is 90 cm, find the diameter of n th dark ring for λ_1 .

Solution : The diameter of n th dark ring due to λ_1 is given by

$$[D_n^2]_{\lambda_1} = 4nR\lambda_1$$

The diameter of $(n+1)$ th dark ring due to λ_2 is given by

$$[D_{n+1}^2]_{\lambda_2} = 4(n+1)R\lambda_2$$

According to the problem

$$[D_n^2]_{\lambda_1} = [D_{n+1}^2]_{\lambda_2}$$

$$\therefore 4nR\lambda_1 = 4(n+1)R\lambda_2$$

$$n\lambda_1 = (n+1)\lambda_2$$

$$\text{or } n = \frac{\lambda_2}{\lambda_1 - \lambda_2} = \frac{4.5 \times 10^{-5}}{(6.0 - 4.5) \times 10^{-5}} = 3$$

$$\begin{aligned}\text{Now } [D_n^2]_{\lambda_1} &= 4 \times 3R\lambda_1 \\ &= 4 \times 3 \times 90 \times 6 \times 10^{-5} \\ [D_n]_{\lambda_1} &= \sqrt{4 \times 3 \times 90 \times 6 \times 10^{-5}} \\ &= 0.254 \text{ cm.}\end{aligned}$$

EXAMPLE 30 In Newton's ring experiment, the diameters of the 4th and 12th dark rings are 0.400 cm and 0.700 cm respectively. Find the diameter of the 20th dark ring.

Solution : We know that

$$D_{n+p}^2 - D_n^2 = 4p\lambda R$$

Here $(n+p) = 12$, $n = 4$, i.e., $p = 12 - 4 = 8$,

$D_{n+p} = 0.700$ cm, and $D_n = 0.400$ cm.

$$\therefore D_{12}^2 - D_4^2 = 4 \times 16 \times \lambda R \quad \dots(1)$$

For 20th and 4th rings

$$\begin{aligned}D_{20}^2 - D_4^2 &= 4 \times 16 \times \lambda R \\ [\because p &= 20 - 4 = 16] \quad \dots(2)\end{aligned}$$

Dividing (2) by (1)

$$\frac{D_{20}^2 - D_4^2}{D_{12}^2 - D_4^2} = \frac{4 \times 16 \times \lambda R}{4 \times 8 \times \lambda R} = 2$$

$$\text{or } \frac{D_{20}^2 - (0.4)^2}{(0.7)^2 - (0.4)^2} = 2 \quad \text{or } \frac{D_{20}^2 - 0.16}{1.1 \times 0.3} = 2$$

$$\text{or } D_{20}^2 - 0.16 = 1.1 \times 0.3 \times 2 = 0.66$$

$$D_{20}^2 = 0.66 + 0.16 = 0.82 \quad \text{or } D_{20} = 0.906 \text{ cm.}$$

EXAMPLE 31 Newton's rings formed by sodium light between a flat glass plate and a convex lens are viewed normally. What will be the order of the dark ring which will have double the diameter of that of 40th dark ring.

Solution : We know that $D_n^2 = 4n\lambda R$

According to the given problem,

$$D_{40}^2 = 4 \times 40 \times \lambda R \quad \dots(1)$$

Let n be the order of dark ring whose diameter is twice that of the diameter of 40th dark ring, i.e., for this dark ring diameter is $(2D_{40})$. Hence

$$(2D_{40})^2 = 4n\lambda R \quad \dots(2)$$

From equations (1) and (2)

$$\frac{(2D_{40})^2}{(D_{40})^2} = \frac{n}{40} \quad \text{or } n = 40 \times 4 = 160$$

EXAMPLE 32 In a Newton's rings experiment the diameter of the 10th ring changes from 1.40 to 1.27 cm when a liquid is introduced between the lens and the plate. Calculate the refractive index of the liquid.

Solution : When the liquid is used, the diameter of the 10th ring is given by

$$(D_{10})^2 = \frac{4 \times 10 \lambda R}{\mu}$$

For air medium $(D_{10})^2 = 4 \times 10 \lambda R$

$$\therefore \mu = \frac{D_{10}^2}{(D'_{10})^2} = \frac{(1.40)^2}{(1.27)^2} = 1.215.$$

EXAMPLE 33 Light containing two wavelengths λ_1 and λ_2 falls normally on a planoconvex lens of radius of curvature R resting on a glass plate. If the n th

dark ring due to λ_1 , coincides with the $(n+1)$ th dark ring due to λ_2 , prove that the radius of n th dark ring of λ_1 is

$$= \sqrt{\left[\frac{\lambda_1 \lambda_2 R}{(\lambda_1 - \lambda_2)} \right]}$$

Solution : We know that the radius of n th dark ring due to λ_1

$$= \sqrt{n \lambda_1 R} \quad \dots(1)$$

The radius of $(n+1)$ th dark ring due to λ_2

$$= \sqrt{(n+1) \lambda_2 R} \quad \dots(2)$$

According to the problem, both are equal, hence

$$r = \sqrt{n \lambda_1 R} = \sqrt{(n+1) \lambda_2 R}$$

$$\text{or } n \lambda_1 R = (n+1) \lambda_2 R$$

$$n = \frac{\lambda_2}{\lambda_1 - \lambda_2} \quad \dots(3)$$

From eqn. (3), substituting the value of n in eq. (1), we have

$$r = \sqrt{n \lambda_1 R} \\ = \sqrt{\left[\frac{\lambda_1 \lambda_2 R}{(\lambda_1 - \lambda_2)} \right]}$$

26.23 MICHELSON INTERFEROMETER

Construction. The apparatus is shown in Fig. 26.27. It consists of two excellent optically plane, highly polished plane mirrors M_1 and M_2 which

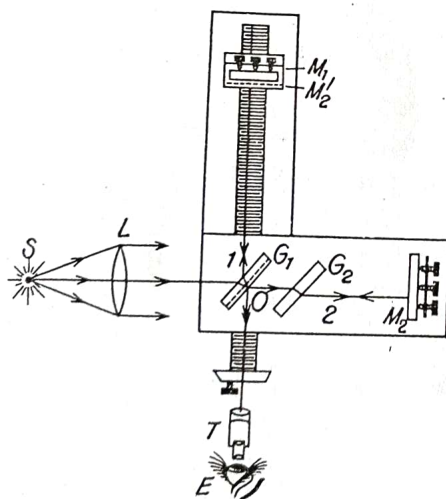


Fig. 26.27

are at right angles to each other. There are two optically flat glass plates. G_1 and G_2 of same thickness and of the same material placed parallel to each other. These plates are also inclined at an

angle 45° with the mirrors M_1 and M_2 . The face of G_1 towards G_2 is semisilvered. The mirror M_1 is mounted on a carriage which can be moved forward or backward. The motion is controlled by a very fine micrometer screw (capable of reading upto 10^{-5} cm). The mirrors M_1 and M_2 are provided with three levelling screws at their backs. With the help of these screws the mirrors can be tilted about horizontal and vertical axes so that they can be made exactly perpendicular to each other. T is a telescope which receives the reflected lights from mirrors M_1 and M_2 .

Working. Light from a monochromatic extended source S after being rendered parallel by a collimating lens L falls on the semi-silvered glass plate G_1 . It is divided into two parts, one being reflected from the semi-silvered surface of G_1 giving rise to ray (1) which travels towards mirror M_1 and other being transmitted giving rise to ray (2) which travels towards mirror M_2 . The two rays fall normally on mirrors M_1 and M_2 respectively and are reflected back along their original paths. The reflected ray again meet at the semi-silvered surface of G_1 and enter a short focus telescope T . The two rays entering the telescope are originally derived from the same single beam, hence they are in a position to produce interference fringes in the field of view of the telescope. A desired path difference can be introduced between the two reflected rays by moving the mirror M_1 .

It can be noted from the figure that ray No. 1 passes through G_1 twice whereas the ray No. 2 does not do so even once. Thus in the absence of glass plate G_2 , the two paths OM_1 and OM_2 are not equal. To equalise the two paths, a glass plate G_2 of same thickness and material as that of G_1 is introduced in the path of ray No. 2. Because of this nature, the glass plate G_2 is called as *compensating plate*. The introduction of G_2 is not essential in monochromatic light but it is a must in case of white light.

Looking in the direction M_1 from E , one observes M_1 and also a virtual image M'_2 of M_1 and M_2 formed in G_1 . Thus the two interfering beams come by reflection from mirror M_1 and the other which is reflected from M_2 functions as if it had been reflected from M'_2 . Hence the Michelson inter-

INTERFERENCE
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When the two mirrors are inclined as shown in Figs. 26.28 (a) and (b), curved fringes with convexity towards the thin edge of the wedge are observed. These fringes are visible up to path differences comparatively much smaller than that in case of circular fringes.

3. Localised white light fringes. When monochromatic light source is replaced by a white source, then coloured and curved localised fringes are obtained (only when the air film has a small thickness). The fringes of zero thickness being again perfectly dark and straight. Other fringes are coloured due to overlapping of various colours. If the film is thick, uniform illumination is observed. White light fringes are useful for the determination of zero path difference, especially in the standardisation of the metre.

26.25 USES OF MICHELSON'S INTERFEROMETER

Michelson's interferometer has been used for a variety of purposes for example (1) in the determination of wavelength of monochromatic source of light (2) to determine the difference between the two neighbouring wavelengths or resolution of the spectral lines, (3) in the determination of refractive index and thickness of various thin transparent materials, (4) for the measurement of the standard meter in terms of the wavelength of light, etc. Out of these uses, some are discussed below :

(1) Determination of wavelength of monochromatic light. First of all the Michelson's interferometer is set for circular fringes with central